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Finite-Pressure Anisotropic Hyperelasticity in FEniCSx: A Phenomenological Approach for Shock Physics

Paul Bouteiller

CEA DAM

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Presenter: Paul Bouteiller (CEA)

We present a finite-strain anisotropic hyperelastic model suited to materials subjected to high-amplitude shock loading, where hydrostatic pressures far exceed deviatoric stresses. The formulation relies on a multiplicative decomposition of the deformation gradient, separating the pressure-induced lattice distortion from the residual isochoric distortion. The Helmholtz free energy is then expanded as a quadratic form in the intermediate Green-Lagrange strain, cleanly decoupling three independently calibrated contributions: the equation of state, the geometric lattice distortion, and the pressure-dependent isochoric stiffness tensor. Each component can be identified directly from DFT calculations, molecular dynamics simulations, or Diamond Anvil Cell experiments. The model naturally covers all crystallographic symmetry classes from cubic to triclinic through a gauge freedom in the isochoric stiffness.

The constitutive law is implemented in the open-source code CHARON (https://github.com/PaulBouteiller/Charon), built on FEniCSx. The total Lagrangian formulation and the UFL automatic differentiation framework allow the free energy to be transcribed almost directly from its mathematical expression, yielding both the stress and the consistent tangent without any additional derivation effort from the user.

Validation is carried out on cubic copper, tetragonal PETN, orthorhombic alpha-RDX. Polycrystal homogenisation on 500-grain meshes generated with NEPER accurately reproduces the pressure dependence of the macroscopic shear modulus.